

Employee Leave Management System on AWS

Volume 1

Project Overview

A serverless leave management application built using AWS Lambda, API Gateway, DynamoDB, Amazon S3, Amazon SNS, IAM and CloudWatch.

Objectives

- Build a production-style serverless application.
- Learn event-driven architecture.
- Practice AWS Free Tier services.
- Create a portfolio-ready cloud project.

Architecture Workflow

Employee → API Gateway → Lambda → DynamoDB

Lambda → Amazon S3 (documents)

Lambda → SNS (email notifications)

CloudWatch stores logs and metrics.

AWS Services

AWS Lambda: Business logic

API Gateway: REST APIs

DynamoDB: Leave data

S3: Leave documents

SNS: Email notifications

IAM: Permissions

CloudWatch: Monitoring

Deployment Steps

Create DynamoDB table

Create S3 bucket

Create SNS topic and email subscription

Create IAM role

Create Lambda

Configure environment variables

Deploy Lambda

Best Practices

Use least-privilege IAM permissions.

Use environment variables instead of hardcoded values.

Enable CloudWatch logging.

Use unique S3 bucket names.